

Sofia Martellozzo

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Google-certified ML Engineer with 2+ years of experience building end-to-end AI systems, from research to production. Specialized in computer vision, multi-agent architectures, and scalable cloud infrastructure on GCP and AWS.

EXPERIENCE

Datwave | *AI Engineer* Sept. 2024 – Present

- Designed and deployed a modular **Computer Vision** platform on GCP spanning object detection and anomaly detection across 3 domains: sports, airport, construction sites and medical.
- Engineered a **Multi-Agent System** using ADK with human-in-the-loop feedback, reducing manual document processing time by automating complex synthesis pipelines.
- Optimised model throughput via **distributed training** across **multi-GPU/multi-VM** environments using the **Accelerate** library and custom model sharding, cutting training time significantly.
- Orchestrated scalable AI workloads on **Kubernetes** (GKE); built secure cloud architectures with **Terraform** and **CI/CD** pipelines, improving deployment reliability.
- Developed an **end-to-end product** using AI to parse and clean architectural drawings, map all safety-relevant elements with HITL validation, compute optimal escape routes, and determine emergency lighting placement.
- Built an **LLM-powered audio analytics pipeline** using **Gemini** and prompt engineering to transcribe and analyse customer support calls, with results visualized in Looker dashboards for business intelligence and decision-making.

ContentWise | *Machine Learning Engineer* Apr. 2024 – July 2024

- Architected an end-to-end **MLOps infrastructure** on AWS for vehicle and license plate recognition, reducing manual pipeline intervention through full automation with ClearML.
- Implemented Infrastructure as Code (IaC) with Terraform, enabling autoscaling and interactive monitoring dashboards for production workloads.
- Integrated an end-to-end pipeline for automatic **license plate extraction (OCR + rule-based ensemble)** and country-of-origin **classification** with a fine-tuned **Swin Transformer** model.
- Integrated **real-time alerting** via Slack and Email, enabling immediate error response in production pipelines and reducing mean time to resolution.

ContentWise | *ML Research Intern (Master Thesis)* Sept. 2023 – Apr. 2024

- Designed and implemented a **Hybrid Search System** on **Weaviate**, combining semantic retrieval (dense **embeddings** + cosine similarity) with keyword search (BM25) for the media domain, improving search relevance over baseline keyword-only approaches.
- Generated a specialised **synthetic dataset** using GPT APIs to simulate complex user queries and responses, enabling robust evaluation without costly manual labelling.
- Evaluated **LoRA** and **Adapter** fine-tuning techniques on multilingual embedding models, selecting the optimal configuration for production deployment.

EDUCATION

Politecnico di Milano Apr. 2024
Master's Degree in Computer Science — Artificial Intelligence 110L

Universidad Politecnica de Madrid Feb.-June 2023
Erasmus+ Program

Politecnico di Milano July 2021
Bachelor's Degree in Computer Engineering

CERTIFICATIONS

Professional Machine Learning Engineer Certification by Google October 2024

SKILLS

ML & AI: PyTorch, HuggingFace, Keras, Transformers (BERT, sBERT, Swin), LoRA, Adapters, ADK
Cloud & MLOps: GCP, AWS, Kubernetes (GKE), Terraform, Docker, ClearML, CI/CD, GitFlow
Data & Databases: Weaviate, Elasticsearch, MongoDB, Neo4j, MySQL, pandas, NumPy
Languages: Python, SQL, Java, C/C++, JavaScript, Terraform, Flutter, $\LaTeX$
Frameworks & Tools: FastAPI, React, Node.js, Git, Kibana, Looker, Cursor, VS Code, GoogleColab